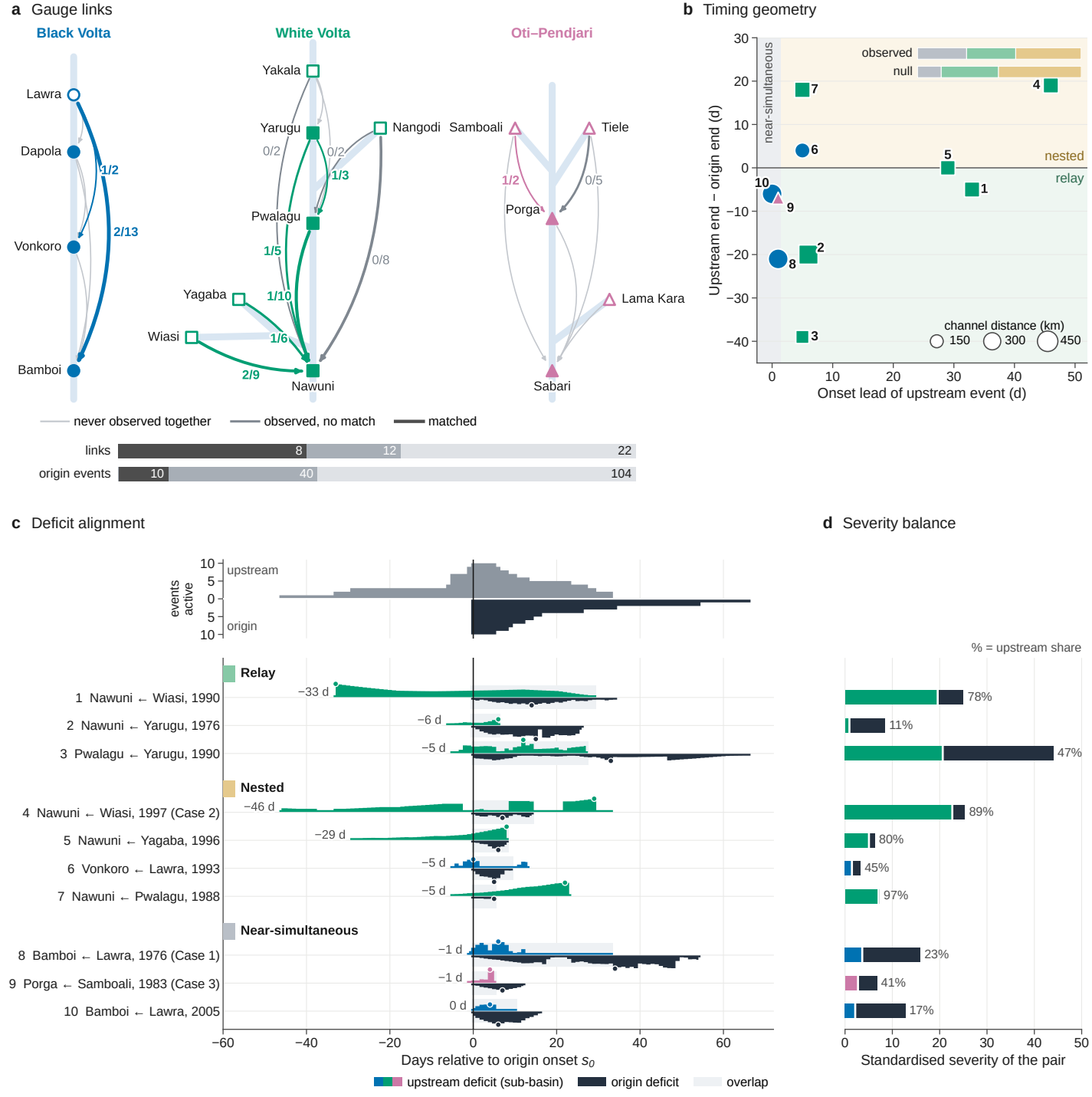

a Gauge links
Black Volta
White Volta
Oti–Pendjari
Lawra
Dapola
Vonkoro
Bamboi
1/2
2/13
Yakala
Yarugu
Nangodi
0/2
0/2
1/3
Pwalagu
1/5
0/8
Yagaba
1/10
Wiasi
1/6
2/9
Nawuni
Samboali
Tiele
1/2
0/5
Porga
Lama Kara
Sabari
never observed together
observed, no match
matched
links
8
12
22
origin events
10
40
104
b Timing geometry
near-simultaneous
observed
null
nested
relay
Upstream end − origin end (d)
Onset lead of upstream event (d)
channel distance (km)
150
300
450
c Deficit alignment
events active
upstream
origin
Relay
1 Nawuni ← Wiasi, 1990
2 Nawuni ← Yarugu, 1976
3 Pwalagu ← Yarugu, 1990
Nested
4 Nawuni ← Wiasi, 1997 (Case 2)
5 Nawuni ← Yagaba, 1996
6 Vonkoro ← Lawra, 1993
7 Nawuni ← Pwalagu, 1988
Near-simultaneous
8 Bamboi ← Lawra, 1976 (Case 1)
9 Porga ← Samboali, 1983 (Case 3)
10 Bamboi ← Lawra, 2005
Days relative to origin onset s_0
upstream deficit (sub-basin)
origin deficit
overlap
d Severity balance
% = upstream share
Standardised severity of the pair